



Network Models

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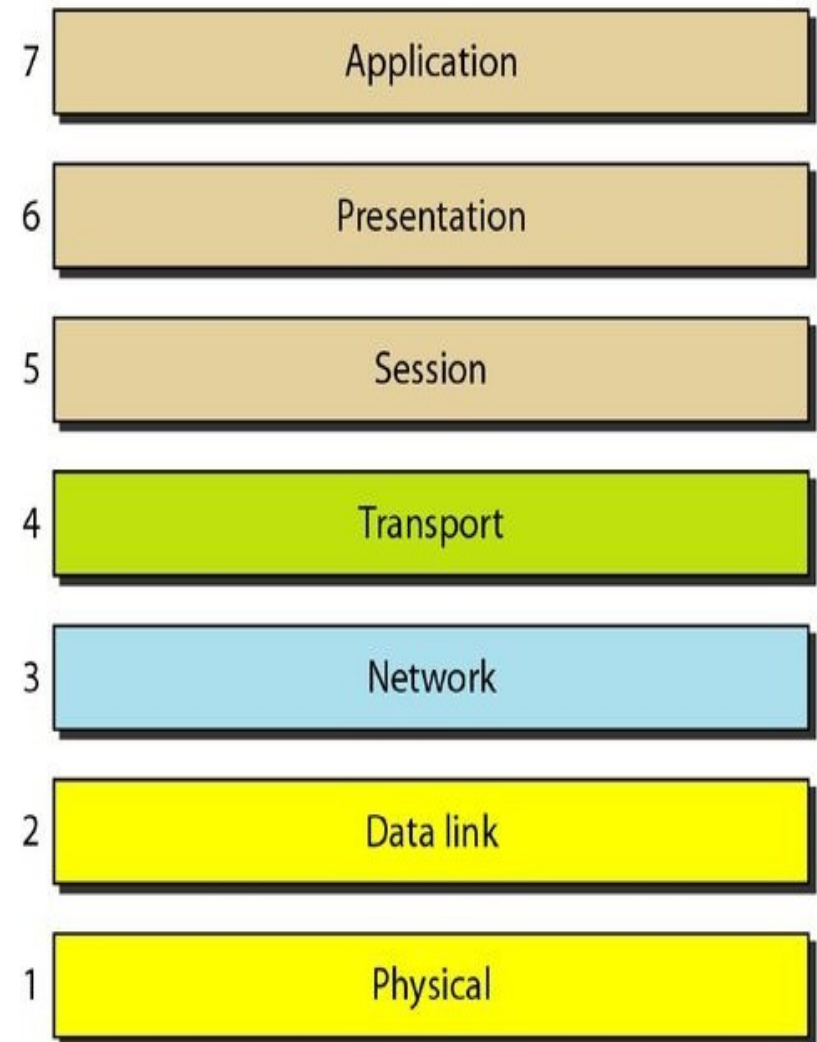
The Layered Model

- To reduce their design complexity, most networks are organized as a stack of **layers** or levels
- Hierarchy
 - The complex task is broken into smaller subtasks
- Services
 - The higher layer uses the services of the lower layer
- The **Open Systems Interconnection (OSI)** model had been dominated data communications and networking literature before 1990
- Later, the **TCPIIP protocol suite** became the dominant commercial architecture because it was used and tested extensively in the Internet
- The OSI model was **never fully** implemented



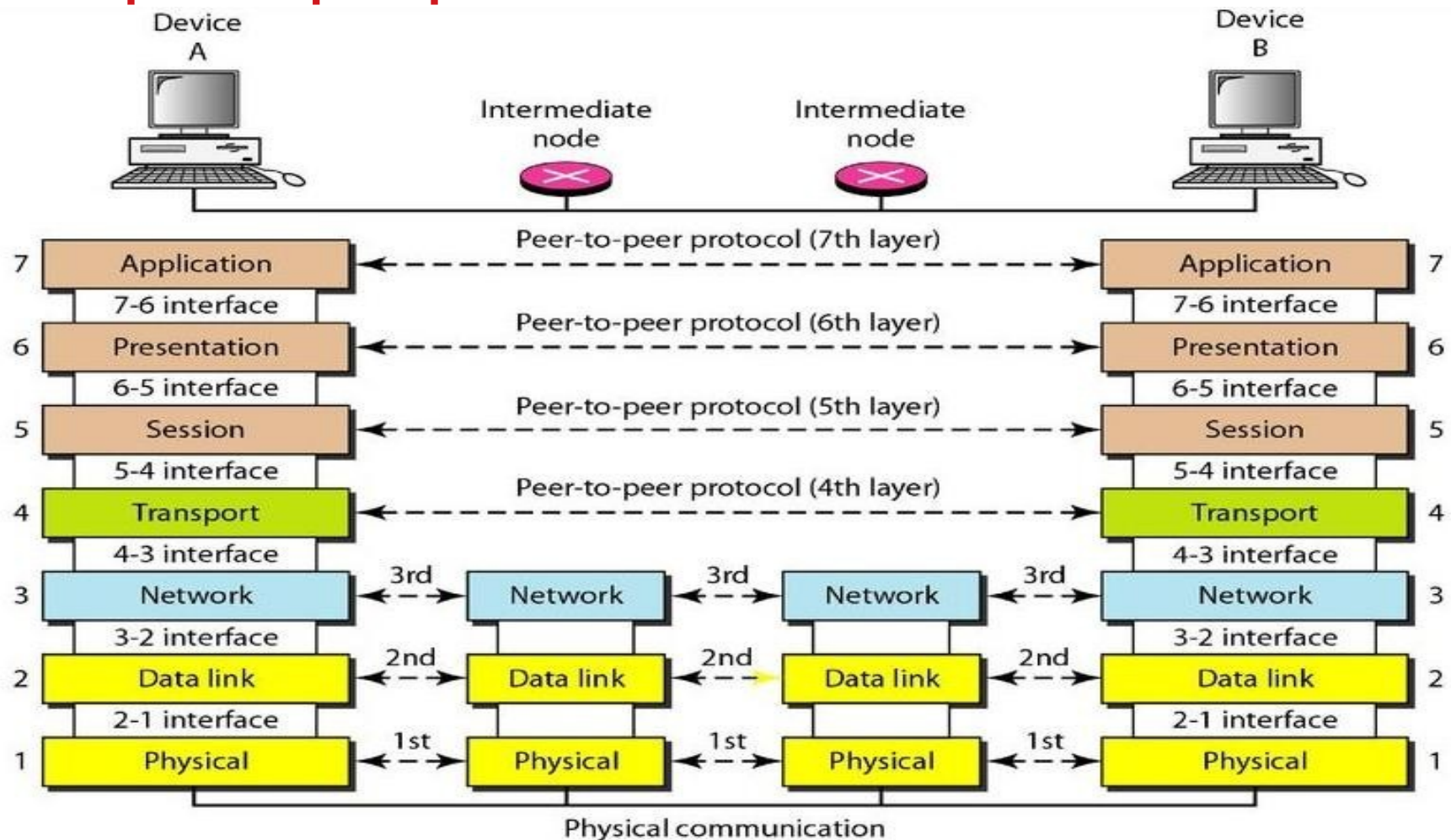
OSI Model

- First introduced in the late 1970s
- A **set of protocols** that allows any two different systems to **communicate** regardless of their **underlying architecture**
- A model for understanding and designing a network architecture that is flexible, robust, and interoperable
- Consists of 7 layers
- Each layer defines how data is treated and goes through different stages while traveling in network from one place to another.
- All layer are used in **end to end systems** but only first three layers used in **intermediate systems** (NODE TO NODE DELIVERY).

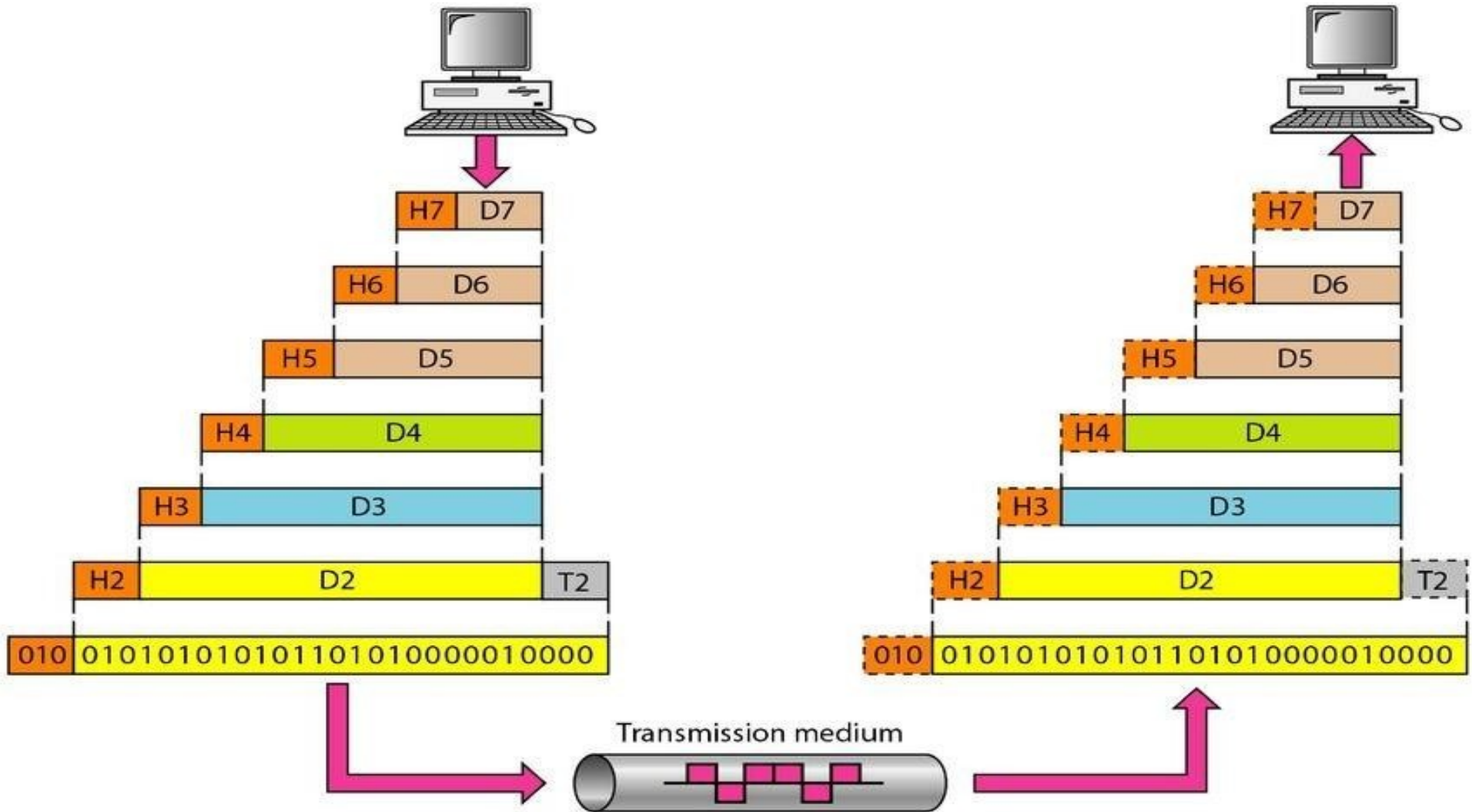


Peer-to-peer communication

- The processes on each machine that communicate at a given layer are called **peer-to-peer processes**

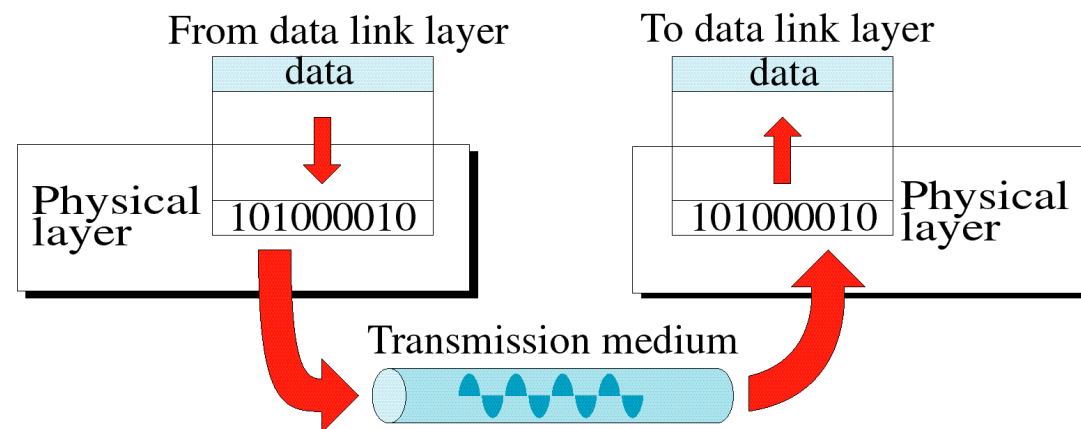


Encapsulation



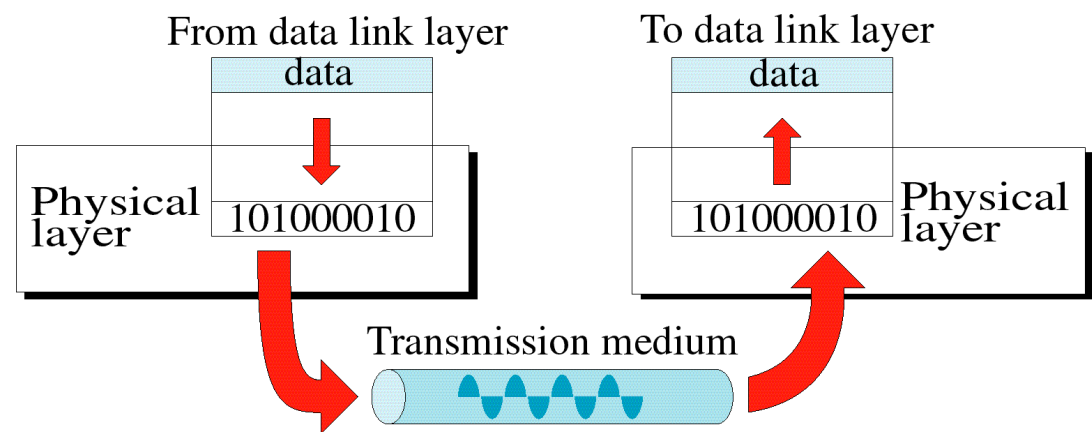
Physical Layer

- The physical layer is responsible for **movements of individual bits from one hop (node) to the next**
- Handles bit stream or binary transmission
- Define physical characteristics of network.
 - Wires
 - Connector
 - Voltages
 - Data rates
 - Synchronization of bits
 - Line configuration, physical topology, and flow mode



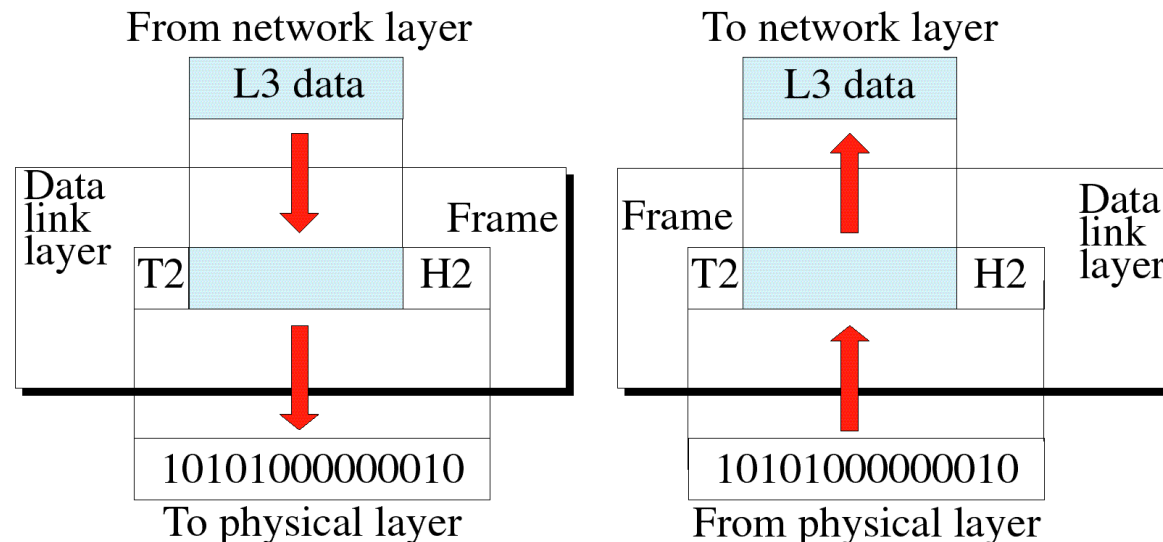
Physical Layer(2)

- Used to maintain, activate and deactivate physical link.
- For receiver it reassembles bits and send to upper layer for frames.
- For Sender it convert frames into bit stream and send on transmission medium.
- Devices working at this layer
 - Hubs
 - Repeaters
 - Cables
 - Modems, etc.,

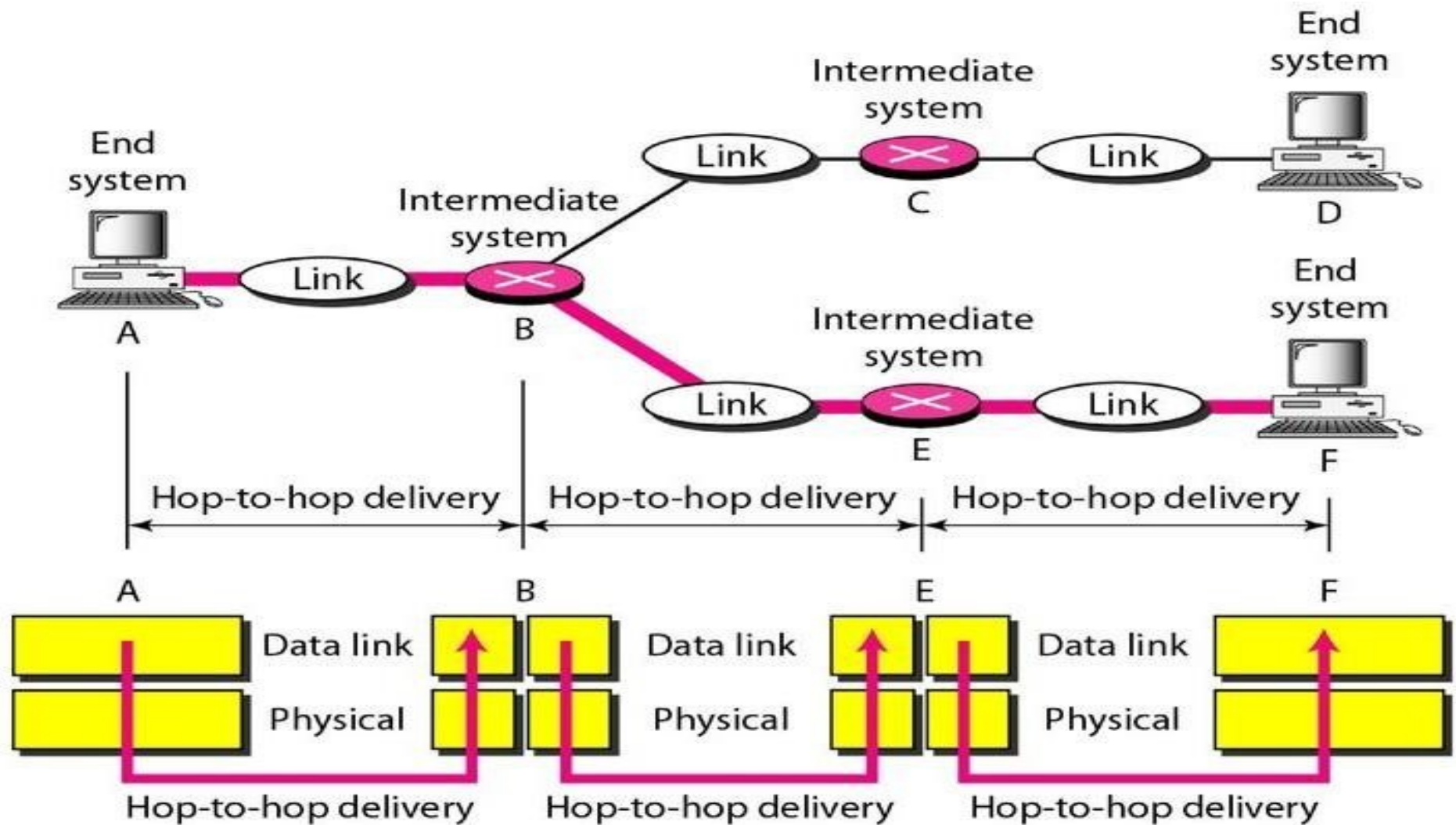


Data Link Layer

- The data link layer transforms the physical layer, a raw transmission facility, to a reliable link
- It makes the physical layer appear error-free to the upper layer (network layer)
- Devices working at this layer
 - Switch
 - Bridge
 - NIC



Hop-to-Hop Delivery in Data Link Layer





Data Link Layer Responsibilities

- **Framing** - The data link layer divides the stream of bits received from the network layer into manageable data units called **frames**.
- **Physical addressing**
 - If frames are to be distributed to different systems on the network, the data link layer adds a header to the frame to define the sender and/or receiver of the frame
 - If the frame is intended for a system outside the sender's network, the receiver address is the address of the device that connects the network to the next one.
- **Flow control** -
 - Speed at which the sender sends and receiver receives the data should be balanced.
- **Error control** -
 - Mechanisms to detect and retransmit damaged or lost frames
 - Also the mechanism to recognize duplicate frames.
 - Error control is achieved through a trailer added to the end of the frame.
- **Access control** -
 - When two or more devices are connected to the same link, layer determines which device has control over the link at any given time.



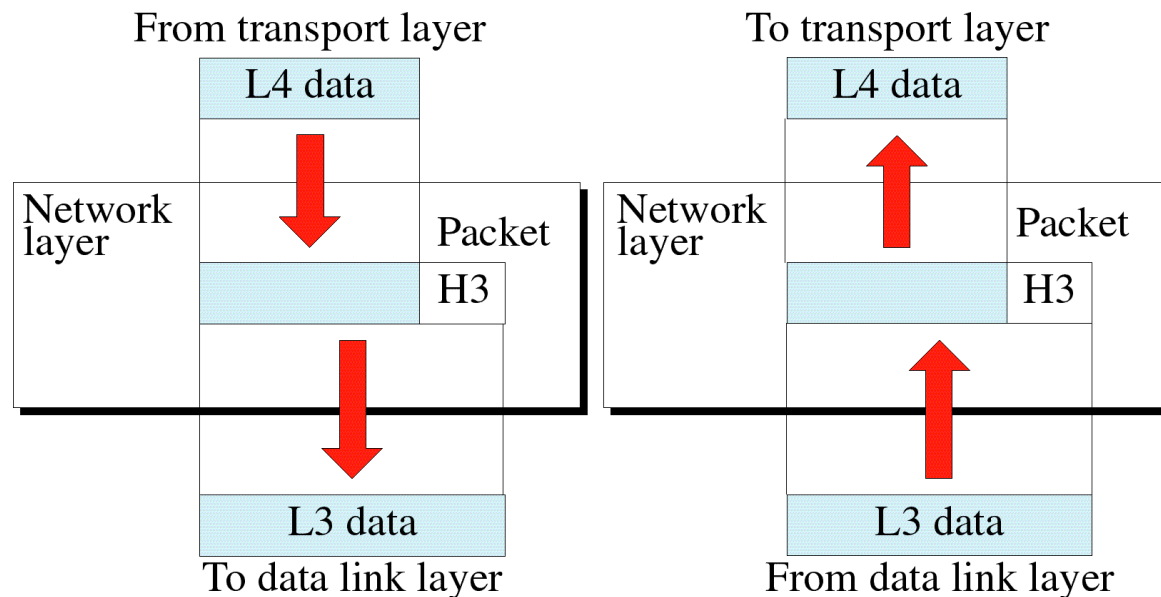
Sub-layers of Data Link Layer

- Logical link layer (LLC)
 - Used for communication with upper layers
 - Error correction
 - Flow control
- Media Access Control (MAC)
 - Access to physical medium
 - Header and trailer

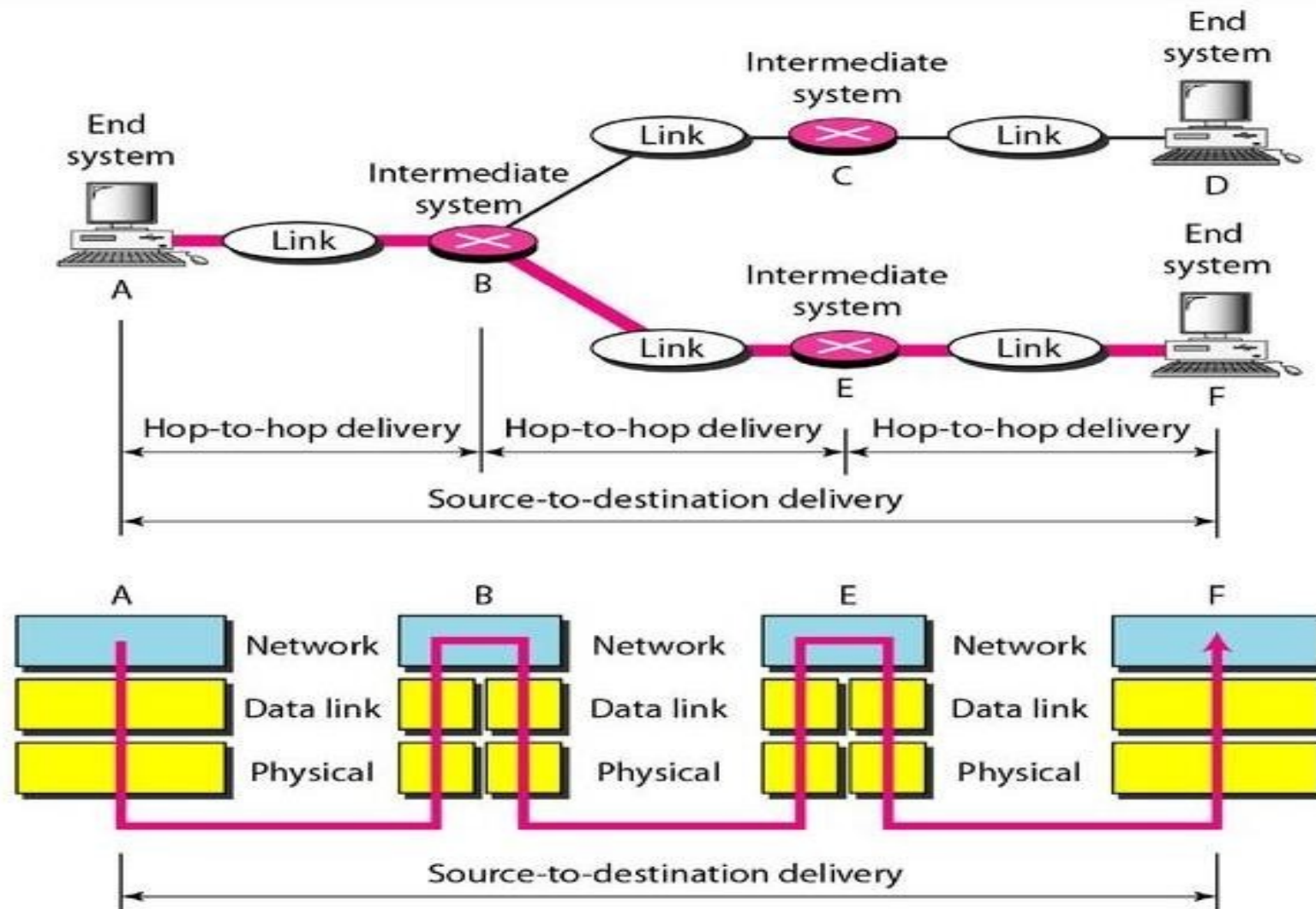


Network Layer

- Data link layer oversees the delivery of the frames between two systems on the **same network** (links)
- It is responsible for the **source-to-destination delivery of a packet**, possibly across **multiple** networks (links)
- Provide switching and routing facilities.
- Determines network address and best path to deliver packets.



Source-to-destination delivery in network layer



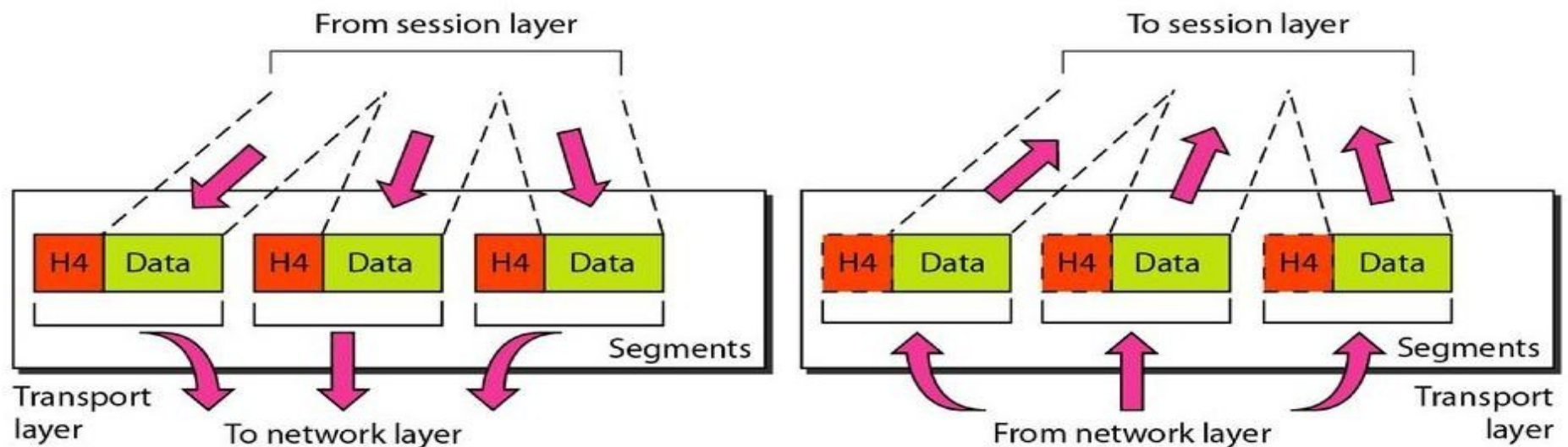


Network Layer Responsibilities

- Logical addressing
 - adds a header that includes the logical addresses of the sender and receiver
- Routing
 - provides route in the independent networks or links to their final destination.

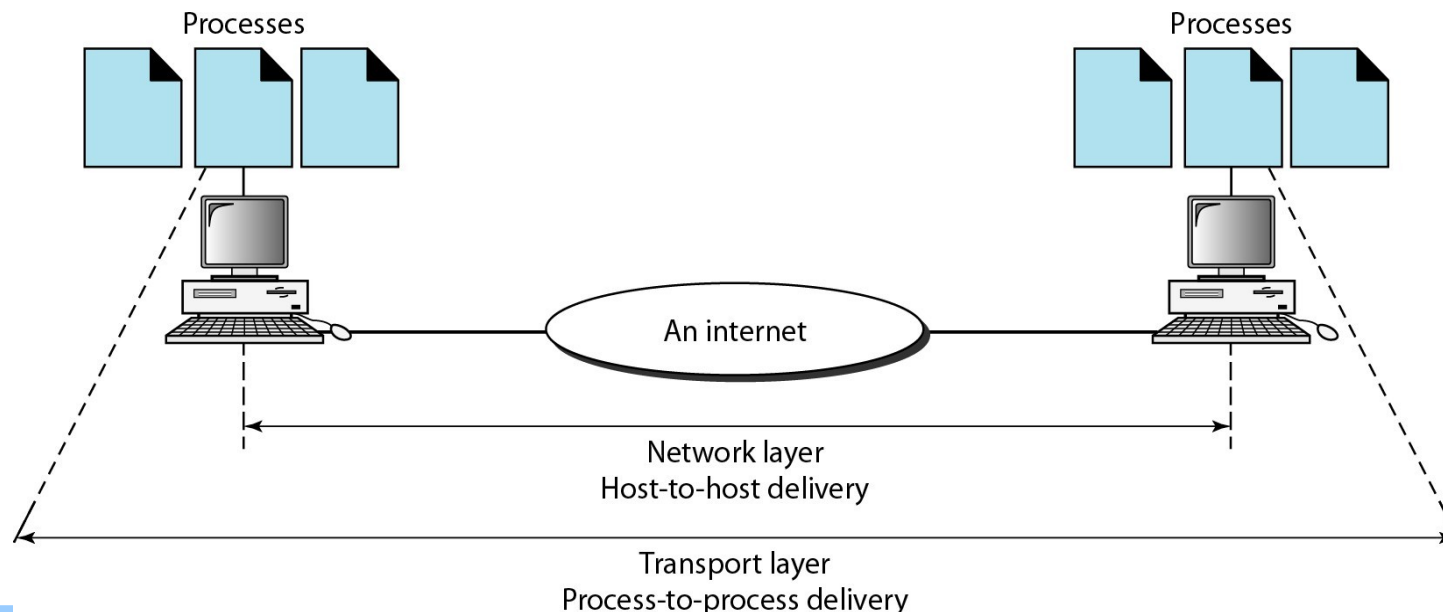
Transport Layer

- The transport layer is responsible for **process-to-process delivery** of the **entire message**
- Network layer does not recognize any relationship between packets, whereas the transport layer ensures that the whole message arrives intact and in order, overseeing both error control and flow control at the source-to-destination level



Transport Layer Responsibilities

- Service-point addressing
 - Source-to-destination delivery means delivery from a process from one computer to process on another
 - Required to include a type of address called a service-point address (or port address)



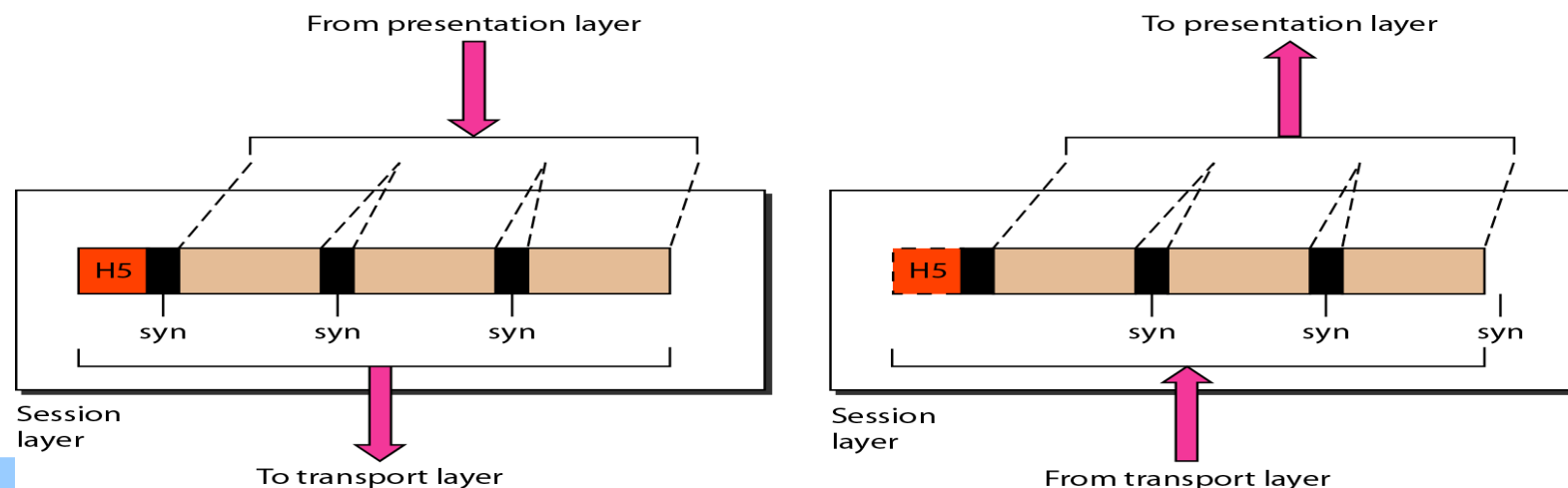


Transport Layer Responsibilities

- Segmentation and reassembly
 - A message is divided into transmittable segments, with each segment containing a sequence number
 - Helps to reassemble the message correctly upon arriving at the destination
- Connection control
 - Either connectionless or connection oriented
- Flow control
 - Like the data link layer, the transport layer is responsible for end-to-end flow control
- Error control
 - Process-to-process error control rather than across a single link
 - Error correction is usually achieved through retransmission

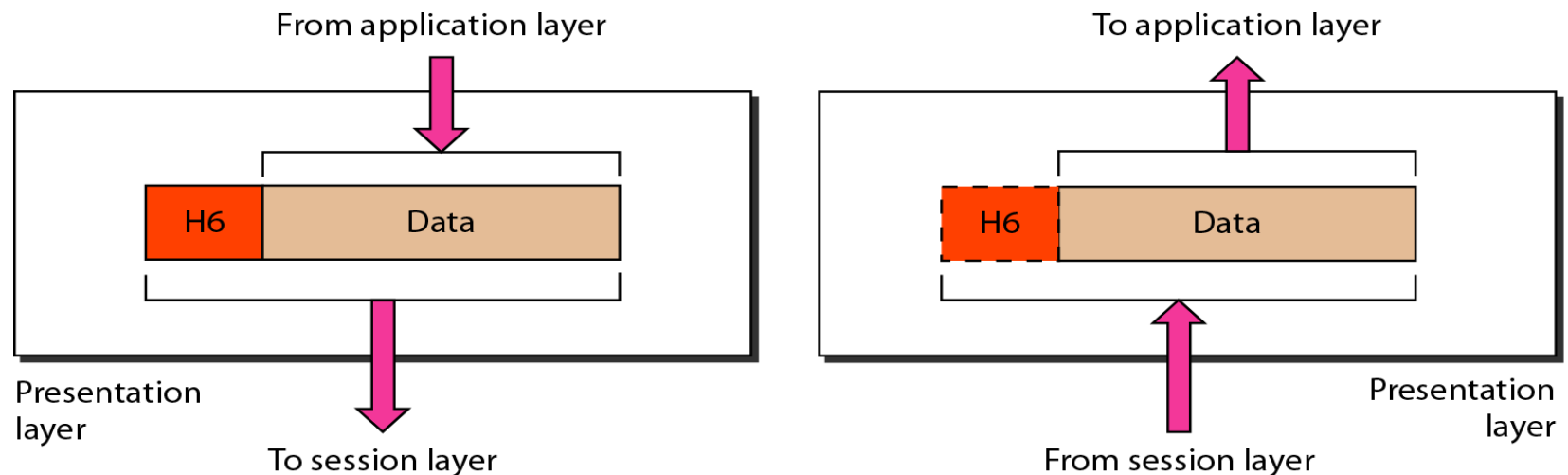
Session Layer

- Used for dialogue control and synchronization purposes.
- Establishes sessions between systems.
- Dialog control:
 - Allows two systems enter into a dialog(either half or full duplex mode).
- Synchronization:
 - Add check points (synchronization points) to stream of data.
 - If session fails only send that data which was not delivered not whole message.



Presentation Layer

- Concerned with the syntax and semantics of the information exchanged between two systems
- Translation
- Encryption
- Compression





Application Layer

- Enables the user, whether human or software, to access the network
- Provides user interfaces and support for services
 - Network virtual terminal
 - File transfer, access, and management
 - Mail services
 - Directory services

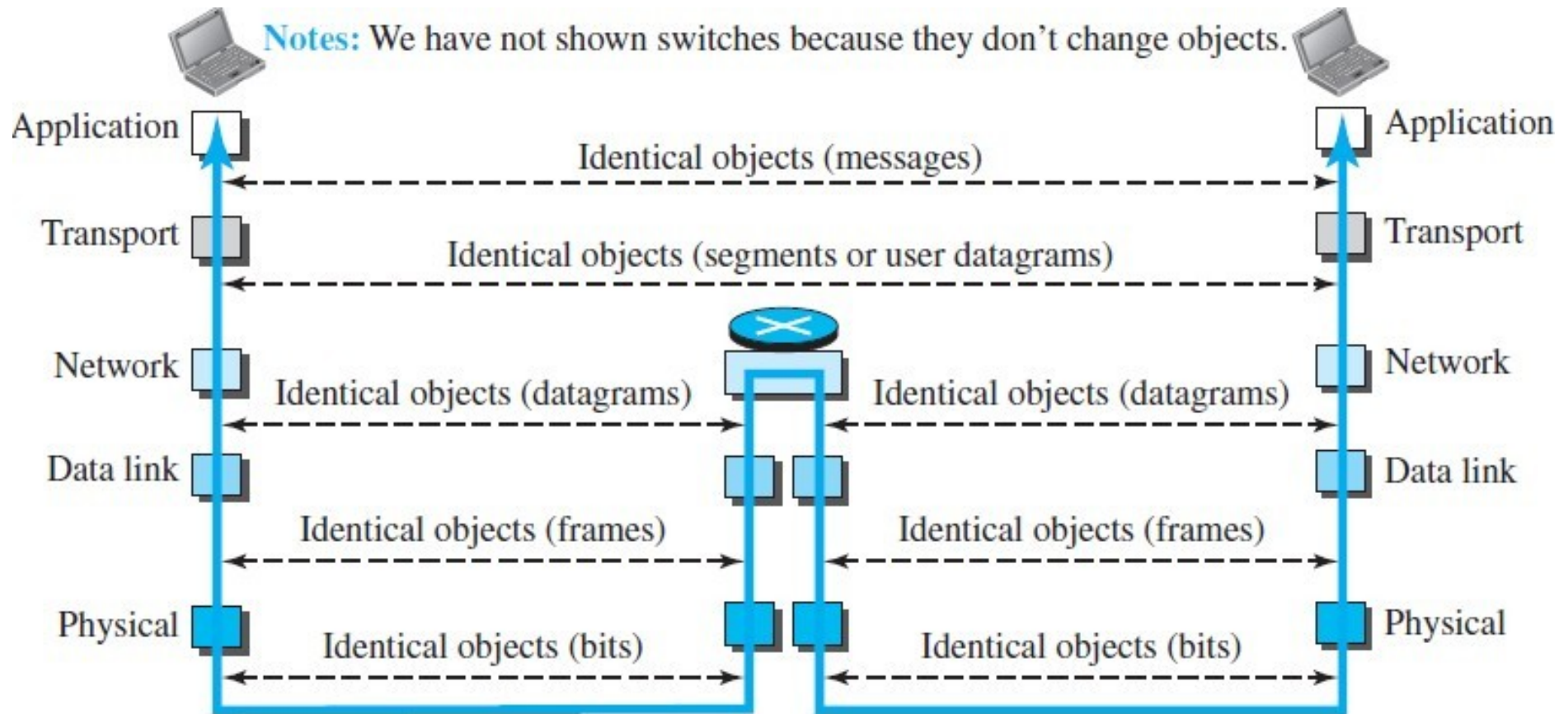


TCP/IP PROTOCOL SUITE

- It is a protocol suite - a set of protocols organized in different layers
- Used in Internet today.
- Five-layer model : physical, data link, network, transport, and application
- It is a hierarchical protocol made up of interactive modules.
- Each module provides a specific functionality.

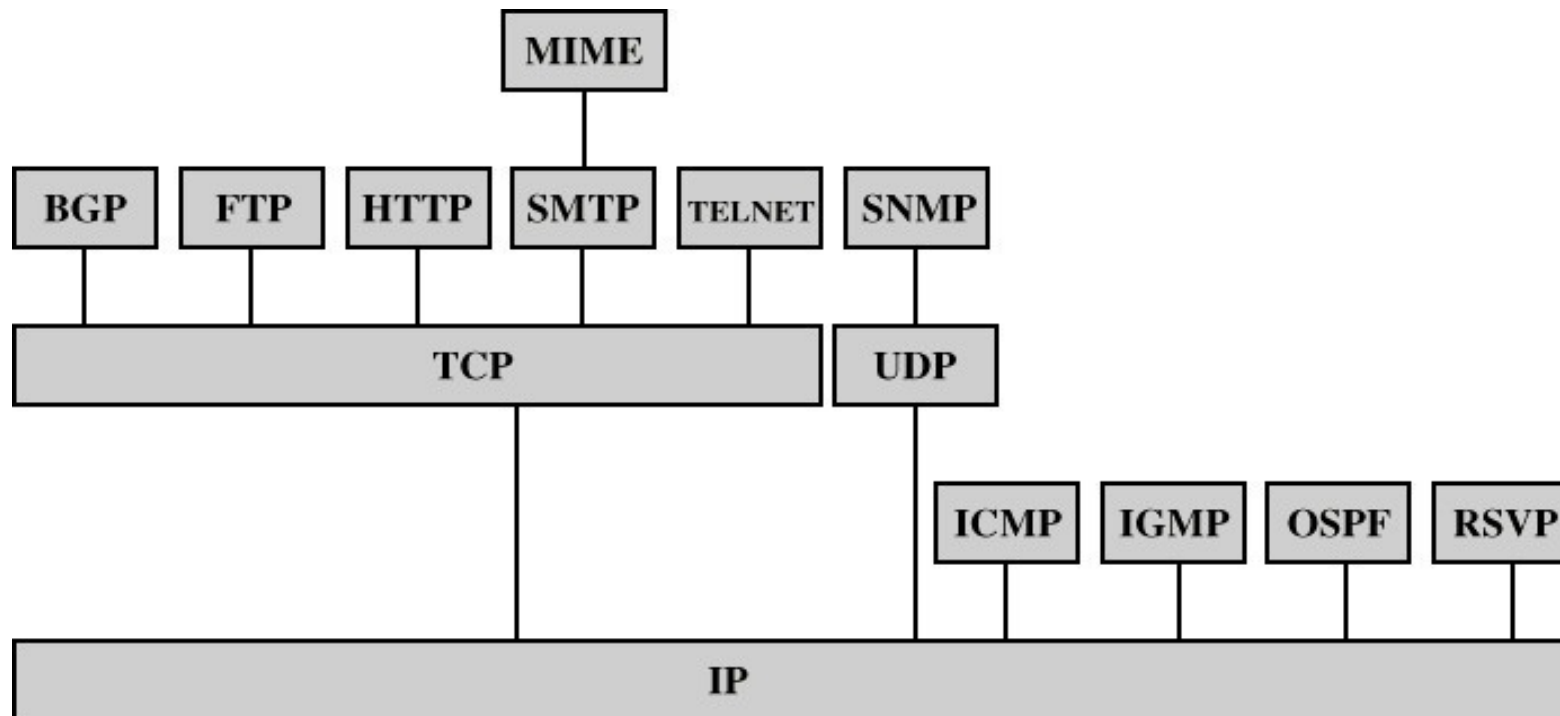
TCP/IP model	Protocols and services	OSI model
Application	HTTP, FTP, Telnet, NTP, DHCP, PING	Application
		Presentation
		Session
Transport	TCP, UDP	Transport
Network	IP, ARP, ICMP, IGMP	Network
Network Interface	Ethernet	Data Link
		Physical

Identical objects





Protocol Stack

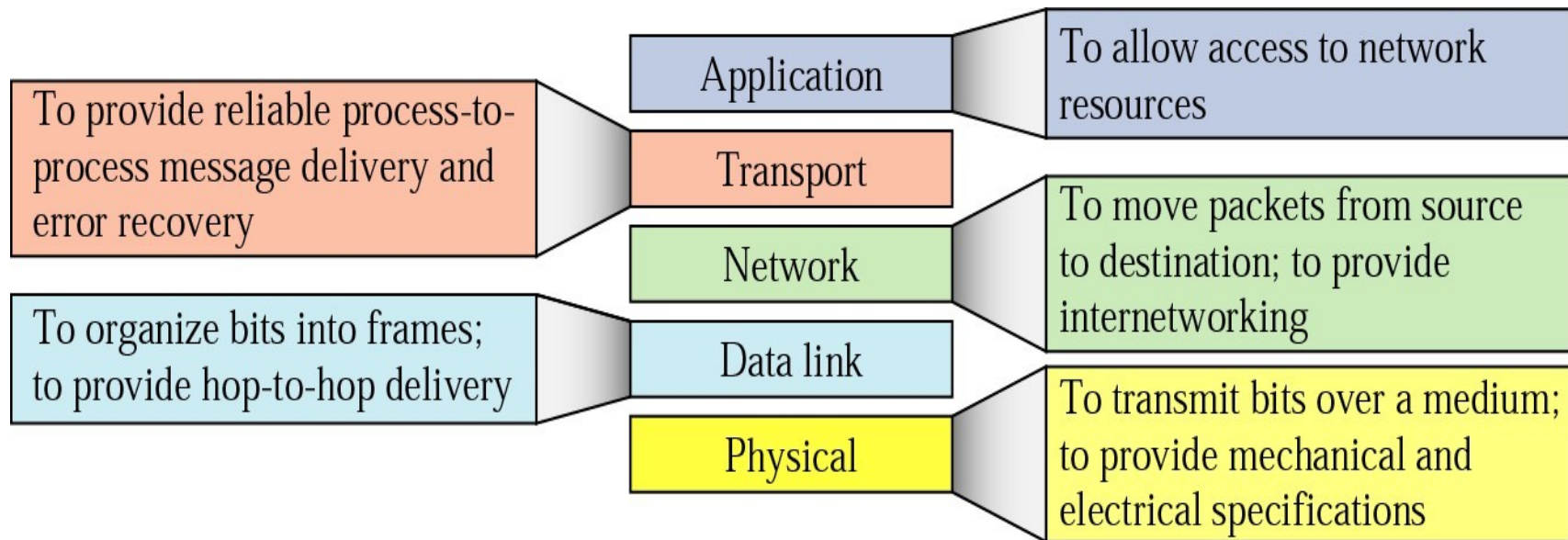


BGP = Border Gateway Protocol
FTP = File Transfer Protocol
HTTP = Hypertext Transfer Protocol
ICMP = Internet Control Message Protocol
IGMP = Internet Group Management Protocol
IP = Internet Protocol
MIME = Multi-Purpose Internet Mail Extension

OSPF = Open Shortest Path First
RSVP = Resource ReSerVation Protocol
SMTP = Simple Mail Transfer Protocol
SNMP = Simple Network Management Protocol
TCP = Transmission Control Protocol
UDP = User Datagram Protocol



Summary





Next : Data and Signal